MIL-C-48115A (PA)
29 March 1976
SUPERSEDING
MIL-C-48115 (MU)
7 September 1972

MILITARY SPECIFICATION

CARTRIDGE, 40MM, HEDP, M430; SPITBACK ASSEMBLY FOR METAL PARTS AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts and the loading, assembling and packing for one type of spitback designated as Spitback Assembly for use in Cartridges 40MM HEDP, M430.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein;

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC: 1310

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-650	- Explosive, Sampling, Inspection Testing

DRAWINGS

PICATINNY ARSENAL

9287862	- Spitback Assembly

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, D. C. 20402. Orders should specify "49 CFR 100-199 (latest revision)").

3. REQUIREMENTS

3.1 Material - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Assembly - The assembly shall comply with all requirement specified on Drawing (Dwg.) 9287862, all associated drawings and with all requirement specified in applicable specifications.

3.3 Workmanship - All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

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3.4 Loading composition - The loading composition at the loading station at the time of loading the spitbacks shall comply with the requirements specified on Drawing (Dwg.) 9287864 and with all requirements specified in applicable specifications.

3.5 Functioning - The spitbacks shall function satisfactorily as evidenced by the RDX functioning high order.

3.6 First Article Inspection - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance provisions - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Liner Spitback	8886352	24
Housing Spitback	9287865	24
Spitback Assembly	9287862	315
Loading Composition, unconsolidated material	9276547	8 oz

4.3.2 Inspections to be Performed - See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection - See MIL-A-48078.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Spitback Assembly and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Liner Spitback (Dwg. 8886352) Examination for Defects)	24	-	4.4.2.1	
	Housing Spitback (Dwg. 9287865) Examination for Defects	24	-	4.4.2.2	
	Spitback Assembly (Dwg. 9287862) Examination for Defects	315	-	4.4.2.4	
	Functioning	315	-	4.5.4	
	Loading Composition				
	Determination of calcium resinate	-	-	4.5.2.1	
	Determination of graphite content	-	-	4.5.2.2	
	Determination of RDX	-	-	4.5.2.3	
	Moisture content	-	-	4.5.3	

NOTES:

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain the following:

a. Metal parts of one interfix lot number from one manufacturer.

b. Loading mixture consisting of batches blended by the same process. Each batch shall be composed of the same ingredients, each of which is to be from one interfix lot from one manufacturer.

4.4.2 Examination - See MIL-A-48078.

a. Sampling Plans - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Liner Spitback		8886352
			NEXT HIGHER ASSEMBLY 9287862

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	See 4.5.1			
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Minimum wall thickness, after forming		0.40%	3.2	Gage
Minor					
201	Spherical radius		0.65%	3.2	Gage
202	Height		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.3	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Housing Spitback		9287865
			NEXT HIGHER ASSEMBLY 9287864

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Smallest outside diameter		0.40%	3.2	Gage
102	Second largest inside diameter		0.40%	3.2	Gage
103	Runout of second largest inside diameter with smallest outside diameter including perpendicularity with flange		0.40%	3.2	Gage
104	Diameter of counterbore		0.40%	3.2	Gage
105	Concentricity of counterbore with second largest inside diameter		0.40%	3.2	Gage
106	Distance from small end to flange		0.40%	3.2	Gage
107	Depth of counterbore		0.40%	3.2	Gage
108	Location small basic diameter		0.40%	3.2	Gage
109	Location large basic diameter		0.40%	3.2	Gage
Minor					
201	Distance from large end to front face of flange		0.65%	3.2	Gage
202	Thickness of flange		0.65%	3.2	Gage
203	Diameter of flange		0.65%	3.2	Gage
204	Body diameter		0.65%	3.2	Visual
205	Large Chamfer on flange missing		0.65%	3.3	Visual
206	Evidence of poor workmanship				

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Spitback Loading Assembly				9287864
					NEXT HIGHER ASSEMBLY
					9287862
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Charge above flush, small end		0.40%	3.2	Gage
102	Charge above flush, liner end		0.40%	3.2	Gage
Minor					
201	Caked explosive on liner seating surface		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.3	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Spitback Assembly		9287862
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Functioning	125	0.40%	3.2	4.5.4
102	Total length		0.40%	3.2	Gage
103	Liner missing		0.40%	3.2	Visual
104	Lacquer missing		0.40%	3.2	Visual
105	Charge above flush		0.40%	3.2	Gage
Minor					
201	Crimp not full 360 degrees		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

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4.4.3 Testing - Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection Equipment - The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test Methods and Procedures

4.5.1 Dimensional control of stamped parts - In place of the normal sampling associated with the Classification of Defects and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.2 Composition

4.5.2.1 Determination of Calcium Resinate - Transfer an accurately weighed portion of approximately 5 grams (gm) of the sample to a tared medium porosity, sintered glass crucible. Extract the specimen with three separate 20 millilitar (ml) portions of chloroform which have been previously saturated with

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RDX, maintaining each portion in contact with the residue *for* approximately five minutes before applying suction. Stir occasionally to break up any lumps and wash down the sides of the crucible with each 20 mi portion of chloroform to insure complete removal of the calcium resinate. Aspirate the residue until the odor of chloroform is no longer detectable, place the crucible and contents in a drying oven at 100 degrees Centigrade (oC) to 105oC for one half hour, cool in a desiccator and weigh. Repeat the above extraction, drying and weighing procedures with individual 20 ml portions of chloroform until the loss in weight on subsequent extractions total less than 0.003 gm. Calculate the loss in weight of the crucible plus contents to percent calcium resinate on a moisture-free basis as follows:

$$\text{Percent calcium resinate} = \frac{100A}{W}$$

Where:

A = loss in weight of the crucible plus contents
W = weight of specimen on a moisture-free basis

4.5.2.2 Determination of the Graphite Content - Extract the residue remaining after calicum resinate (See 4.5.2.1) has been removed with two 20-ml portions of dimethyl formamide, maintaining each portion in contact with the residue for approximately five minutes with intermittent stirring before applying suction. Wash the crucible and contents with a minimum of three 20-ml portions of acetone, maintaining each portion in contact with the residue for approximately one minute with. continuous stirring before applying suction until the following qualitative test for RDX in the acetone washings is negative. Transfer the acetone filtrate to a 25 ml beaker, evaporate to dryness on a steam bath and add several drops of a solution of chromotrophic acid in 36 normal (N) sulfuric acid. The formation of a reddish-brown color as compared to a pink color obtained by a blank prepared in an identical manner, indicates the presence of RDX. Aspirate the residue until the odor of acetone is no longer detectable, place the crucible and contents in a drying over at 100 degrees to 105 degrees C for one half hour, cool in a desiccator and weigh. Calculate the weight of the residue in the crucible to percentage of lubricant in the sample on a moisture-free basis as follows:

$$\text{Percent lubricant} = \frac{100\ A}{W}$$

Where:

A = increase in weight of empty tared crucible
W = weight of sample on a moisture-free basis

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4.5.2.3 Determination of RDX - The percent RDX shall be determined by subtracting the percent calcium resinate and lubricant from 100 percent.

4.5.3 Moisture content of composition - Content shall be determined using Karl Fischer method in accordance with Standard MIL-STD-650, Method 101.4 with methanol as the special solvent.

4.5.4 Functioning - The spitback assembly shall be assembled to a HE loaded cartridge for which it is standard. The cartridge shall be fired from a launcher for which it is standard to impact on 3/4" plywood sheet located 200 feet from the launcher. Observation shall be made for proper functioning (See 3.5). If fuze fails to function, the test sample shall be disregarded. If more than four (4) fuze duds occur, replacement samples must be fired.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

5.1.1 Level C - The liner spitback and housing spitback shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing -

5.2.1 Level C - The liner spitback and housing spitback preserved and packaged as specified in 5.1.1 shall be packed for shipment in accordance with MIL-sTD-1169.

5.3 Marking - Marking of the carton shall comply with MIL-STD-1169 and Code of Federal Regulations, Title 49, Parts 100-199.

5.4 Shipping - When components from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in trnasit.

6. NOTES

6.1 Intended Use - The assembly covered by the specification is intended for use in the Fuze, PIBD, M549.

6.2 Ordering Data - See MIL-A-48078.

6.3 Submission of Inspection Equipment Designs for Approval - See MIL-A-48078. Submit equipment designs as required to Commander, Attn: SARPA-QA-T, Picatinny Arsenal, Dover, NJ 07801

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6.4 Submission of Results of Contractor Conducted Examinations and Tests - Data shall be submitted in accordance with data item DI-R-1721 on the DD Form 1423 for the contract.

6.5 Submission of Test Data - In addition to the normal distribution of record when the spitback assembly is procured by the US Army Armament Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, Picatinny Arsenal, Attn: SARPA-QA-A-S.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1310-A087

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